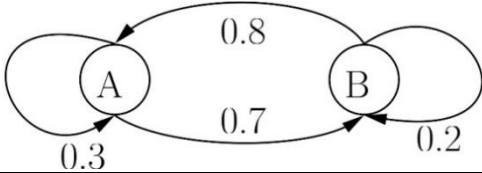




Date: 30/06/2025	Re- CIE	Max. Marks:100
Semester: VI	UG	Duration: 2 Hrs. (2pm-4pm)
Course Title: Mathematical Modelling		Course Code: MA266TEU

Department of Mathematics

	Quiz			
SL.No	Questions	M	BT	CO
1	The angle of projectile to attain the maximum horizontal distance should be _____ the maximum height should be _____.	2	L1	2
2	A hot object at 120°C is placed in a room with temperature 30°C. If the cooling constant is $k = 0$, what is the temperature after 5 minutes?	2	L1	1
3	In the long run, the proportion of time that the given Markov chain will be in state A is _____ and state B is _____. 	2	L2	3
4	Define probability vector. Check whether the vector $(\frac{1}{2}, \frac{1}{4}, \frac{1}{4}, 0)$ is a probability vector or not.	2	L2	2
5	The growth of an organism follows the law $\frac{dx}{dt} = \frac{1}{13}x$. If initially there were 100 of them, then after 13 years, their approximate number will be _____ and their doubling period is approximately _____ years.	2	L1	1
SL. No	Test	M	BT	CO
1.	A drug is induced in a patient's bloodstream at regular intervals of duration T each. Simultaneously the drug disappears at a rate proportional to the concentration $c(t)$ of the drug present at any time t. Determine the differential equation governing $c(t)$, solve it and interpret the results.	10	L3	2
2.	Write the systematic 12-step procedure for tackling a problem through the use of mathematical modelling.	10	L1	1
3.a	Describe the concept of a simple compartment model with an example.	05	L2	2



3.b	The territory of a sales man consists 3 cities. A, B, C. he never sells in the same city on successive days. If he sells in city A, then the next day he sells in B. However, if he sells either in B or C, then the next day he is twice likely to sell in city A as in the other city. How often does he sell in each of the cities in steady state?	05	L2	3
4.a	Using the limiting behaviour of the homogeneous Markov chain, find the steady state probability of the chain given by the TPM $P = \begin{bmatrix} 0.1 & 0.6 & 0.3 \\ 0.5 & 0.1 & 0.4 \\ 0.1 & 0.2 & 0.7 \end{bmatrix}$	05	L2	3
4. b	Explain linear and decay model with an example.	05	L1	1
5	A gambler has ₹2. He bets ₹1 at a time and wins ₹1 with probability $\frac{1}{2}$. He stops playing if he ends up with no money or wins ₹4. a. What is the TPM of the related Markov chain? b. What is the probability that he has lost his money at the end of 5 plays? c. What is the probability that the game lasts more than 7 plays?	10	L3	4



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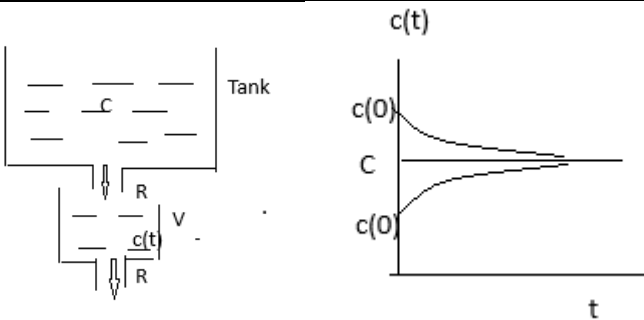
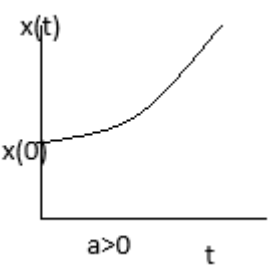
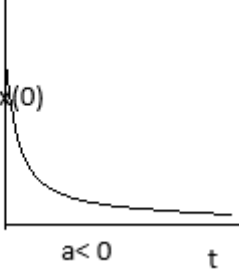
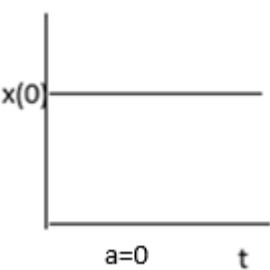
SCHEME & SOLUTIONS

S No	Solutions with Scheme	Marks
	Part-A	
1.	$\frac{\pi}{4}, \frac{\pi}{2}$	1+1
2.	$T(t) = T_a + (T_0 - T_a)e^{-kt}$, $T(5) = 100.092^\circ\text{C}$	1+1
3.	$\pi_A = 158 \approx 0.5333$ $\pi_B = 157 \approx 0.4667$, $\text{TPM} = \begin{matrix} 0.3 & 0.7 \\ 0.8 & 0.2 \end{matrix}$	1+1
4.	Probability vector	1+1
5.	271, 9	2

Q.No	Part-B	Marks
1	let a dose D of a medicine be given to a patient at regular intervals of duration T each. The medicine also disappears from the system at a rate proportional to $c(t)$, the concentration of the medicine in the blood stream Then by continuity principle $V \frac{dc}{dt} = -kc$ Integrating we get $c(t) = De^{-\frac{k}{V}t} \quad 0 \leq t < T \quad \text{----- (4)}$ At time $T(t \geq T)$, the residue of the first dose is $De^{-\frac{k}{V}T}$ and now another dose D is given so that we get $c(t) = De^{-\frac{k}{V}t} + De^{-\frac{k}{V}(t-T)} \quad T \leq t < 2T$ The first term gives the residual of the first dose and the second term gives the residual of the second dose. Proceeding in this way after n doses we get $c(t) = De^{-\frac{k}{V}t} + De^{-\frac{k}{V}(t-T)} + De^{-\frac{k}{V}(t-2T)} \pm \dots \pm De^{-\frac{k}{V}(t-(n-1)T)}$ $= De^{-\frac{k}{V}t} \left[1 + e^{\frac{k}{V}T} + e^{\frac{2k}{V}T} + \dots + e^{(n-1)\frac{k}{V}T} \right] = De^{-\frac{k}{V}t} \frac{\left(e^{\frac{nk}{V}T} - 1 \right)}{e^{\frac{k}{V}T} - 1}, \quad (n-1)T \leq t < nT$ Just before the next dose $c(nT - 0) = De^{-\frac{nk}{V}T} \frac{\left(e^{\frac{nk}{V}T} - 1 \right)}{e^{\frac{k}{V}T} - 1} = \frac{D(1 - e^{-\frac{nk}{V}T})}{e^{\frac{k}{V}T} - 1}$ When next dose is given	2+2+ 2+2+2



	$c(nT + 0) = D e^{-\frac{nk_r}{V}T} \frac{(e^{\frac{k_r}{V}T} - 1)}{e^{\frac{k_r}{V}T} - 1} + D = \frac{D(e^{\frac{k_r}{V}T} - e^{-\frac{k_r}{V}T})}{e^{\frac{k_r}{V}T} - 1}$ As $n \rightarrow \infty$ we have $D \frac{e^{\frac{k_r}{V}T} - 0}{e^{\frac{k_r}{V}T} - 1}, \frac{D}{\frac{e^{\frac{k_r}{V}T} - 1}{e^{\frac{k_r}{V}T}}} = \frac{D}{1 - e^{-\frac{k_r}{V}T}}$. The concentration never exceeds $\frac{D}{1 - e^{-\frac{k_r}{V}T}}$. In each interval concentration decreases. In any interval, the concentration is maximum at the beginning and that maximum concentration keep increasing interval by interval but will be below $\frac{D}{1 - e^{-\frac{k_r}{V}T}}$. The minimum value in an interval occurs at the end of each interval. This also keep increasing but lies below $\frac{D}{e^{\frac{k_r}{V}T} - 1}$. The concentration curve is piecewise continuous and has points of discontinuity at $T, 2T, 3T, \dots$ By injecting glucose on penicillin in blood and fitting curve (4) to the data, we can estimate the value of k and V. In particular this gives a method for finding the volume of blood in the human body.	
2	twelve-point procedure for solving problems through mathematical modeling. Be clear about the real-world situation to be investigated. Find all its essential characteristics relevant to the situation and the aspects that can be ignored. (ie of minimal relevance) Think about all the physical, chemical, biological, social, economic laws that may be relevant to the situation. Formulate the problem into problem language (PL) Think about all the variable $x_1, x_2, \dots, x_n$ and the parameters $a_1, a_2, \dots, a_m$ involved. Classify them into known and unknown ones. Choose the most appropriate model and translate the problem suitably into Mathematical Language (ML) Think of all possible ways of solving the problem. The methods may be analytical (Preferably), numerical or simulation. If a reasonable change in the assumptions makes analytical solution possible, investigate the possibility. Make an error analysis of the method used. If the error is not within acceptable limits, change the method of solution. Translate the final solution into PL Compare the predictions with available observations or data. If the agreement is good, accept the model else change the assumptions and approximations in the light of the discrepancies observed and proceed as before. Continue the process till a satisfactory model is obtained. Deduce the conclusions from the model and test these conclusions against easier data and additional data that may be collected and see if the agreement still continues to be good.	1+2 +2+2 +1+2
3a	Let a vessel contain a volume V of a solution with concentration $c(t)$ of a substance at time t Let a solution with constant concentration C in an overhead tank enter the vessel at a constant rate R and after mixing thoroughly with the solution in the vessel, let the mixture with concentration $c(t)$ leave the vessel at the same rate R, so that the volume of the solution in the vessel remains same V	2+2+1

	 Using the principle of continuity we get $V [c(t+\Delta t) - c(t)] = RC\Delta t - Rc(t) \Delta t$ Dividing by Δt and taking $\Delta t \rightarrow 0$ $V \frac{dc}{dt} = RC - Rc, \frac{dc}{dt} = \frac{R}{V} C - \frac{R}{V} c = (C-c) \frac{R}{V}, \log(C-c) = -\frac{R}{V} t + c_1$ $C-c = Ke^{-\frac{R}{V}t}, c(t) = C - Ke^{-\frac{R}{V}t}, c(0) = C - K, K = C - c(0)$ $\therefore c(t) = C - (C - c(0))e^{-\frac{R}{V}t} = c(0)e^{-\frac{R}{V}t} + C(1 - e^{-\frac{R}{V}t})$ As $t \rightarrow \infty$ $c(t) \rightarrow C$, so that ultimately the vessel has the same concentration as the overhead tank. Also if $C > c(0)$ the concentration in the vessel increases to C and $C < c(0)$, the concentration in the vessel decreases to C.	
3b	$\text{TPM} = \begin{bmatrix} 0 & 1 & 0 \\ \frac{2}{3} & 0 & 1/3 \\ 2/3 & 1/3 & 0 \end{bmatrix}, \pi = \frac{8}{20}, \frac{9}{20}, \frac{3}{20}$	1+2 +2
4a	$\pi = [19/94, 24/94, 51/94] \text{ (or } \pi = [19/94, 12/47, 51/94])$	+2 +2+1
4b	Let $x(t)$ be the population size at time t and let b and d be the birth and death rates i.e, the number of individuals born or died per individual per unit time. In time interval $(t, t + \Delta t)$, the number of births and deaths would be $bx\Delta t$ and $dx\Delta t$ $x(t + \Delta t) = x(t) + bx\Delta t - dx\Delta t, \quad = x(t) + (b - d)x\Delta t, x(t + \Delta t) - x(t) = (b - d)x\Delta t \text{ -----(1)}$ Dividing both sides by Δt and taking limit as $\Delta t \rightarrow 0$ (1) becomes $\frac{dx}{dt} = (b - d)x = ax \text{ where } a = b - d, \text{ i.e., } (D - a)x = 0, x = c_1 e^{at}, t=0 \quad x(0) = c_1, \therefore x(t) = x(0)e^{at}$ This implies population grows exponentially if $a > 0$ and decays exponentially if $a < 0$ and remains constant if $a = 0$.    (i) $a > 0$ then let T be the time in which the population will become double its present size $x(t) = 2x(0), \Rightarrow 2x(0) = x(0)e^{aT}, \Rightarrow \log_e 2 = aT, \quad T = \frac{1}{a} \log_e 2$	+2 +2+1



T is called the doubling period of the population and it may be noted that this doubling period is independent of $x(0)$. It depends on only on a and is such that greater the value of a , small is the doubling period.

(ii) If $a < 0$, the population will become half its present size in time, say T'

$$x(t) = \frac{1}{2}|x(t)| \Rightarrow \frac{1}{2}x(0) = x(0)e^{aT'} \Rightarrow \frac{1}{2} = e^{aT'} \Rightarrow \ln\left(\frac{1}{2}\right) = aT', \therefore T' = \frac{1}{a}\ln\left(\frac{1}{2}\right)$$

T' is called the half life (period) of the population. It is also independent of $x(0)$ and it decreases as the excess of death rate over birth rate increases.

Probability and Queueing Theory

$$P^{(2)} = P^{(1)} \cdot P = \left(\frac{1}{4}, 0, \frac{1}{2}, 0, \frac{1}{4}, 0, 0 \right)$$

$$P^{(3)} = P^{(2)} \cdot P = \left(\frac{1}{4}, \frac{1}{4}, 0, \frac{3}{8}, 0, \frac{1}{8}, 0 \right)$$

$$P^{(4)} = P^{(3)} \cdot P = \left(\frac{3}{8}, 0, \frac{5}{16}, 0, \frac{1}{4}, 0, \frac{1}{16} \right)$$

$$P^{(5)} = P^{(4)} \cdot P = \left(\frac{3}{8}, \frac{5}{32}, 0, \frac{9}{32}, 0, \frac{1}{8}, \frac{1}{16} \right)$$

(ii) After the fifth play, the probability of having ₹ 0 = $\frac{3}{8}$, (i.e. he loses ₹ 2).

$$P^{(6)} = P^{(5)} \cdot P = \left(\frac{29}{64}, 0, \frac{14}{64}, 0, \frac{13}{64}, 0, \frac{1}{8} \right)$$

$$(iii) P^{(7)} = P^{(6)} \cdot P = \left(\frac{29}{64}, 0, \frac{14}{128}, 0, \frac{27}{128}, 0, \frac{13}{128}, \frac{1}{8} \right)$$

∴ The game lasts more than 7 plays is (he stops playing only when he has ₹ 0 or ₹ 6)

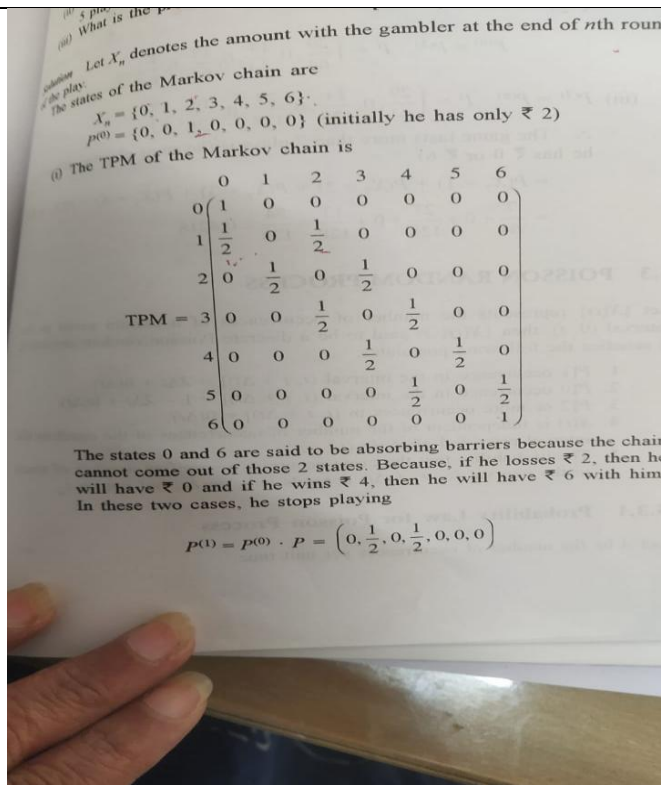
$$\begin{aligned} &= P(X_7 = 1) + P(X_7 = 2) + P(X_7 = 3) + P(X_7 = 4) + P(X_7 = 5) \\ &= \frac{7}{64} + 0 + \frac{27}{128} + 0 + \frac{13}{128} = \frac{54}{128} = 0.4218 \end{aligned}$$

3. POISSON RANDOM PROCESS

Let $\{X(t)\}$ represents the number of occurrences of a certain event in the interval $(0, t)$, then $\{X(t)\}$ is said to be a discrete Poisson random process if it satisfies the following postulates:

1. $P[1 \text{ occurrence in the interval } (t, t + \Delta t)] = \lambda \Delta t + o(\Delta t)$.
2. $P[0 \text{ occurrence in the interval } (t, t + \Delta t)] = 1 - \lambda \Delta t + o(\Delta t)$.
3. $P[2 \text{ or more occurrences in } (t, t + \Delta t)] = o(\Delta t)$.
4. $X(t)$ is independent of the number of occurrences of the event in the interval prior and after $(0, t)$.
5. The probability that the event occurs in a specified number of times in the interval $(t_1, t_1 + \Delta t)$ depends only on Δt , but not on t_1 .

5.3.1 Probability Law for Poisson



SCRUTINY & EVALUATION OF CIE QUESTION PAPER

Date: 02/05/2025	Test - 1	Max. Marks:100
Semester: VI	UG	Duration: 2 Hrs. (2pm-4pm)
Course Title: Mathematical Modelling		Course Code: MA266TEU

Declaration by the Course handling faculties:

As a teaching faculty of the course: **Mathematical Modelling** with code: **MA266TEU**

We hereby confirm that the question paper with Scheme and Solutions is thoroughly reviewed and we



ensure that it adheres to the following criteria:

Sl. No	Criteria	Yes/No
1	The question paper adequately covers the prescribed syllabus contents.	
2	The question paper is in line with the recommended pattern, taking into consideration the structure and format suitable for the Continuous Internal Evaluation.	
3	The question paper is designed to align with the Revised Bloom's Taxonomy, encompassing various levels of cognitive skills such as remembering, understanding, applying, analysing, evaluating, and creating.	
4	The question paper is aligned with the defined course outcomes, ensuring that it effectively assesses the knowledge and skills acquired during the course.	
5	Course handling faculty (as applicable) who are responsible for preparing the question paper, Scheme and Solutions have unanimously agreed to utilize this Question paper for conducting the Continuous Internal Evaluation.	
6	The Question paper, Scheme and complete Solutions have been submitted to the Test coordinators within the designated time-frame to ensure the smooth conduction of Continuous Internal Evaluations	

Course handling Faculties:

Name:

Signature

1. Dr. Y. Sailaja

2. Dr. Nivya Muchikel

To be filled by the Scrutinizer:

Sl. No	Rubrics	Points	
		Max	Awarded
1	Timely submission of the question paper along with the scheme & solution	10	
2	Heterogeneous nature of QP with respect to BTs and Cos	10	
3	Format with proper entry of all particulars including test, course name, code, date, max marks, BT CO table, efficient use of paper (proper spacing, figures)	10	
4	No handwritten data or diagrams, and uniform fonts throughout	10	
5	Scheme & complete Solutions in the format	10	
	Total points	50	



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Bengaluru - 560059, Karnataka, India

Any other comments by the scrutinizer :

Note: Course coordinators to obtain scrutinizer's acceptance by incorporating all suggestions from scrutiny into the final versions of QP, Scheme, and Solutions.

All corrections suggested by the scrutinizer are incorporated and both the copies are re-submitted	Signature of Course coordinator
Accepted/Rejected	Signature of Scrutinizer (Name:)

Signature of HOD